

Orbit SDG user journey

National landcover tasking tool · Global pre-computed SDG indicator metrics

S0 · Who you are	Two audiences before two products. A member state agency needs a map of its own country; a custodian agency publishes an indicator for all of them. The same platform, different things.
S1 · National tasking, and where labels come from	The national path. The tool is indicator-agnostic, so the fork here is not which indicator but where the labels come from — a file, or imagery the user interprets.
N1 · What you are mapping	No indicator list. Label is categorical or continuous, Orbit never thresholds it for you, and the 10 metre rule is an attestation the user ticks.
N2 · Upload and columns	Latitude, longitude and label required; date, position accuracy and site id optional. Coordinate reference and columns are detected, not asked.
N3 · Harmonise and declare absence	Absence is declared, never inferred, and this is the one failure no later check can see. A presence-only survey reads as a confident sample.
N4 · Label readiness	One count gates: labels in the smallest class. Coverage is a readout that never blocks, because the extent is already reported only where the labels are.
A1 · Sampling design	The annotation branch, and where a presence-only upload lands. Whether it works at all is decided per country by the imagery: only the United States has an open high-resolution orthophoto registered today.

A2 · Annotate	The same gate as N4, as progress. Adding labels can only improve the state, never worsen it.
N5 · Partition and run	Both branches converge. Block hold-out only, seed on screen, and one run control: the embedding year. Everything else is fixed and recorded.
N6 · Run console	Lines append, never replace. The console is a filter over the run record, so the two cannot disagree.
N7 · Held-out accuracy	Median with a 10th to 90th band over 10 block-bootstrap draws. Categorical run shown; a continuous run reports R squared and root mean square error.
N8 · Extent and its error	Where the tool stops. A stratified reference-based estimate with an interval, reported only over blocks that hold reference points.
N9 · Export and run record	Class probability is the deliverable. No hard-class raster ships. Tiles are keyed by task id, and the run record is the chain of custody.
F · Blocked	Two things stop a run: no absence class, and not enough labels in a class. The layout does not change between them.

UNOPS SDG GeoAI Tasking Tool

Indicate which type of organisation you are, and what you need.



MEMBER STATE AGENCY

I need a land cover map for my own country

- You bringField points with coordinates, or imagery you interpret here
- You getA national map, the area of the class, and its error
- ScopeOne country, any land characteristic, any year 2017 to 2024
- OwnerYour agency. Orbit forms no opinion about the target

Worked example · Irish peat soils, from 9,914 survey points to a national extent

National landcover tasking tool



CUSTODIAN AGENCY

I need pre-computed global metrics for a specific SDG indicator

- You bringNothing
- You getPublished values for every country, on one frozen method
- ScopeThe indicators Orbit computes, comparable across countries and years
- OwnerOrbit publishes the method and the values

Worked example · Indicators 6.6.1, 11.3.1, 15.1.1, 15.3.1 and 2.4.1

Global pre-computed SDG metrics

The two are separate products. The national tool is indicator-agnostic: it maps whatever land characteristic an agency describes and hands back the extent with its error, without knowing or asking which indicator that feeds. Indicator choice lives entirely on the custodian side, where the method is frozen and published.

National landcover tasking tool

One country, any land characteristic. You keep the result.

Where do your labels come from?



I have field data

In	CSV or GeoPackage with class, latitude, longitude
Out	Landcover map, area, accuracy
Time	Minutes

Upload a file



I will interpret imagery

In	Nothing
Out	The same, from a sample you draw and interpret here
Time	Hours to days, readiness fills as you go

Start annotating

N1 · What you are mapping

No indicator list. Label is categorical or continuous, Orbit never thresholds it for you, and the 10 metre rule is an attestation the user ticks.

What are you mapping?

Country

Ireland



Land characteristic

Peat soil

Label type

☒ Categorical variable

marsh, wetland, bog

☐ Continuous variable

0 to 100

Embedding year

2018



- ☒ **The thing I am mapping is larger than 10 metres across.**
Features smaller than the 10 metre pixel are not detectable with this method. Do not use this tool to map objects or land characteristics below 10 metres.

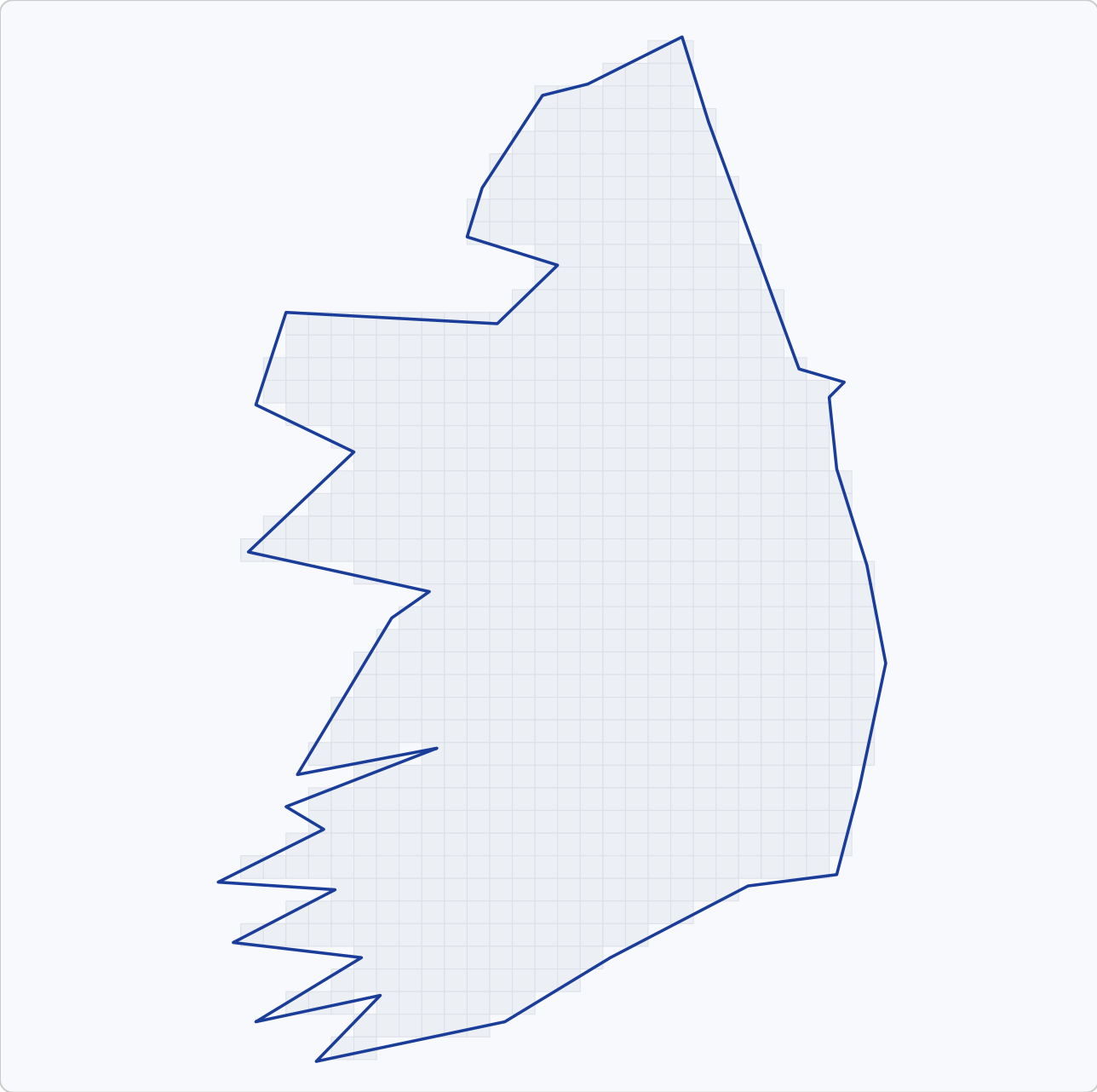
What you intend to use it for, optional

SDG 6.6.1, vegetated wetland extent

Recorded. Not used by the model.

10 kilometre block grid

859 blocks over the Republic



Block, no labels yet

EPSG:2157 · schematic

N2 · Upload and columns

Latitude, longitude and label required; date, position accuracy and site id optional. Coordinate reference and columns are detected, not asked.

Upload your labels



Drop a CSV or GeoPackage

Choose a file

✓ tellus_shallow_topsoil_A.csv · 9,921 rows · 5 columns · 1.4 MB

✓ Coordinate reference detected: EPSG:4326, WGS 84 geographic. From the value ranges and the column names.

Change

Columns, detected Change any of them

Latitude Required

Latitude

▼

Detected

Longitude Required

Longitude

▼

Detected

Label, categorical Required

PEAT_CLASS

▼

Detected, 2 distinct values

Observation date Optional

SAMPLE_DATE

▼

Detected, 2011 to 2023

Position accuracy Optional

Not in this file

▼

No column found, nothing filtered

Site identifier Optional

SAMPLE_ID

▼

Detected

Preview, as the tool reads it

Site	Position	Label	Date	Accuracy
TL-04412	54.2013, -8.4471	Peat	2013-06-11	not supplied
TL-04413	54.1988, -8.4102	Not peat	2013-06-11	not supplied
TL-04414	54.2044, -8.3760	Not peat	2013-06-12	not supplied
TL-04415	54.1902, -8.3391	Peat	2013-06-12	not supplied
TL-04416	54.1877, -8.3055	Not peat	2013-06-12	not supplied

Continue

Back

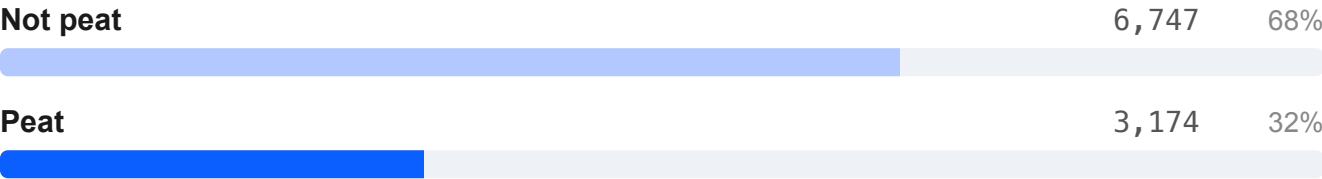
N3 · Harmonise and declare absence

Absence is declared, never inferred, and this is the one failure no later check can see. A presence-only survey reads as a confident sample.

Harmonise the labels

Values in the file

PEAT_CLASS, 9,921 points, 2 distinct values



Map to target classes

☒ Peat 3,174 points Positive class

☐ Not peat 6,747 points Negative class

Does this dataset contain absence?

The one question no later check can answer for you. A presence-only survey with 50,000 points reads as a confident sample and is unusable, because nothing downstream reveals the missing half.

☒ Yes, Not peat is the absence class
6,747 points

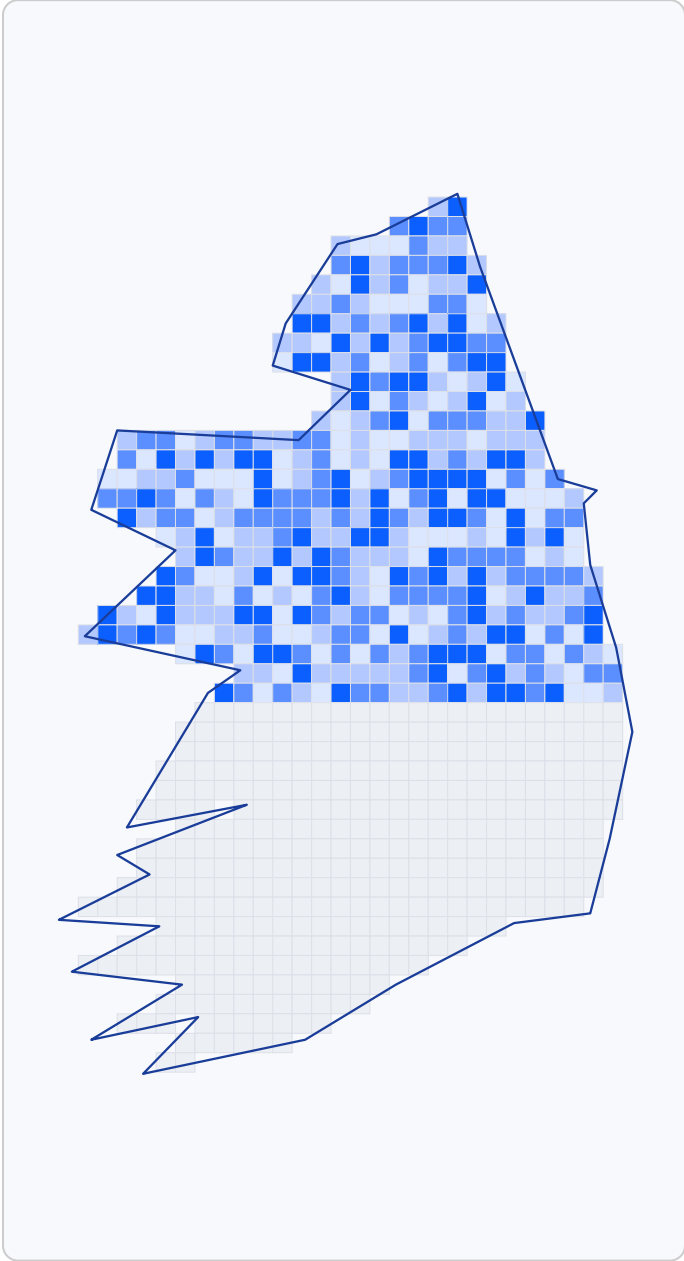
☐ No, this upload is presence only
You will be asked to add absences next

N4 · Label readiness

One count gates: labels in the smallest class. Coverage is a readout that never blocks, because the extent is already reported only where the labels are.

Labels on the 10 km grid

449 of 859



Labels per block ■ ■ ■ ■ ■ None

Label readiness

Your labels match the recommendation

3,191 labels in the smallest class vs 400

You have at least as many labels of each class as the recommendation. **It does not mean your map will reach that accuracy — not every map does.** A fifth case, Spanish forest inventory plots, never matched its national forest map at the 40 percent canopy cut, at any number of plots; adding plots did not close the gap, and the study does not establish what would. **Real agreement statistics come only from a full run, scored against your own held-out labels.** Treat this as a preliminary guide to how much to collect and where.

Per class

The smallest class decides. A verdict is never shown without this table.

Class	Labels	Blocks holding it	
Peat	3,191	370	MATCHES
Not peat	6,723	429	MATCHES

Where you get an answer

READOUT

You will get an answer for **449 of the 859 blocks** in the country. Coverage does not affect the verdict.

Cleaning applied

READOUT

6,865 outside the Republic, dropped · 155 within 500 m of the line, named by id · 0 near-duplicates at 10 m

Not checked here

READOUT

Whether the survey contains absence. That is asked outright on the previous screen, because no count can reveal it.

Calibrated on four national cases — Ireland peat, Brazil forest, United States rangeland and a seven-class global set — at the number of labels where each matched the published map it was tested against. A fifth case, Spanish forest inventory plots against MFE25, never did at the 40 percent canopy cut, at any number of plots. Adding plots did not close the gap, and the study does not establish what would. At the 20 percent cut it does reach it. **A rough guide, not a guarantee.**

Continue to partition

Back

Only *not enough labels* stops a run. Coverage never does.

A1 · Sampling design

The annotation branch, and where a presence-only upload lands. Whether it works at all is decided per country by the imagery: only the United States has an open high-resolution orthophoto registered today.

Sampling design

Mode

Full sample

Absence only

Absence mode is where a presence-only upload lands.

Country

Kenya

Land characteristic

Woody vegetation

Classes

Woody

Not woody

+ Add

Sample locations

800

Allocation

Simple random over bloc

Imagery basemap

None registered for Keny

Imagery year

2023

This branch cannot run here yet

Buffer from your uploaded plots

1 km

Absence mode only

A sample the tool draws is a probability sample, so the estimate is design-based.

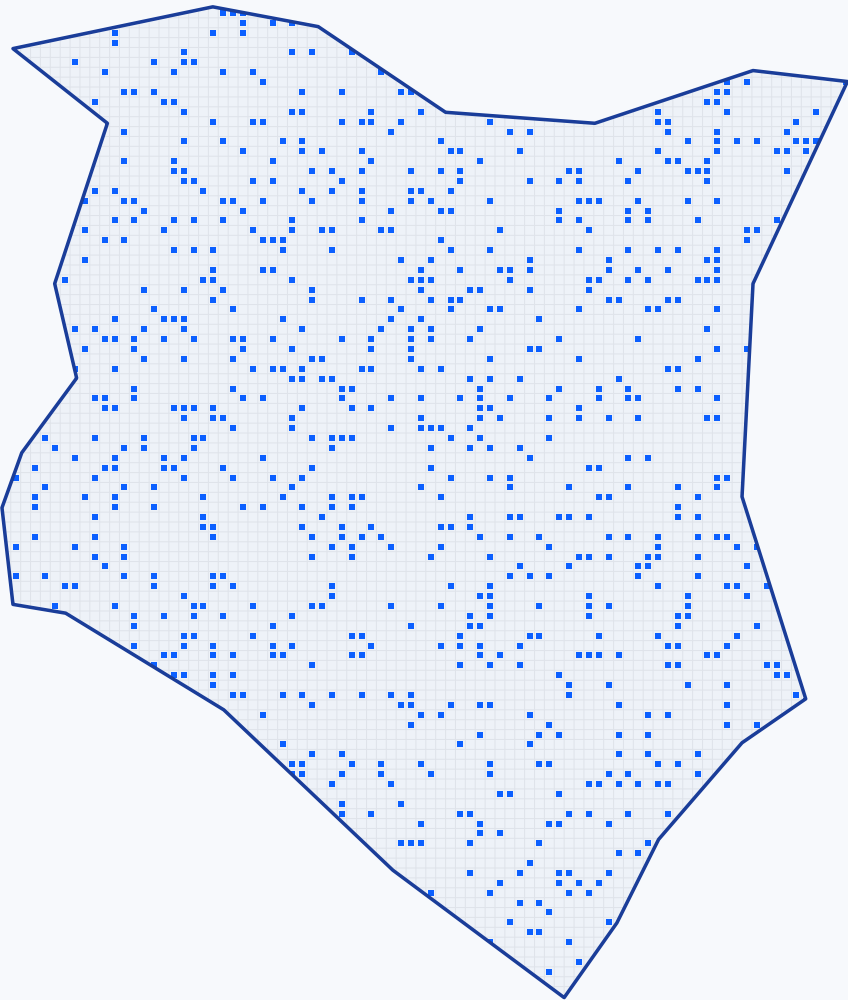
Imagery decides whether this branch works, country by country.
Open high-resolution orthophotos are national, not global — only the United States has one registered today. Sentinel-2 is not a fallback: at 10 m the interpreter and the classifier see the same data.

Start annotating

Back

Where you will be sent

800 locations in 5,473 blocks



Sample location 10 km block

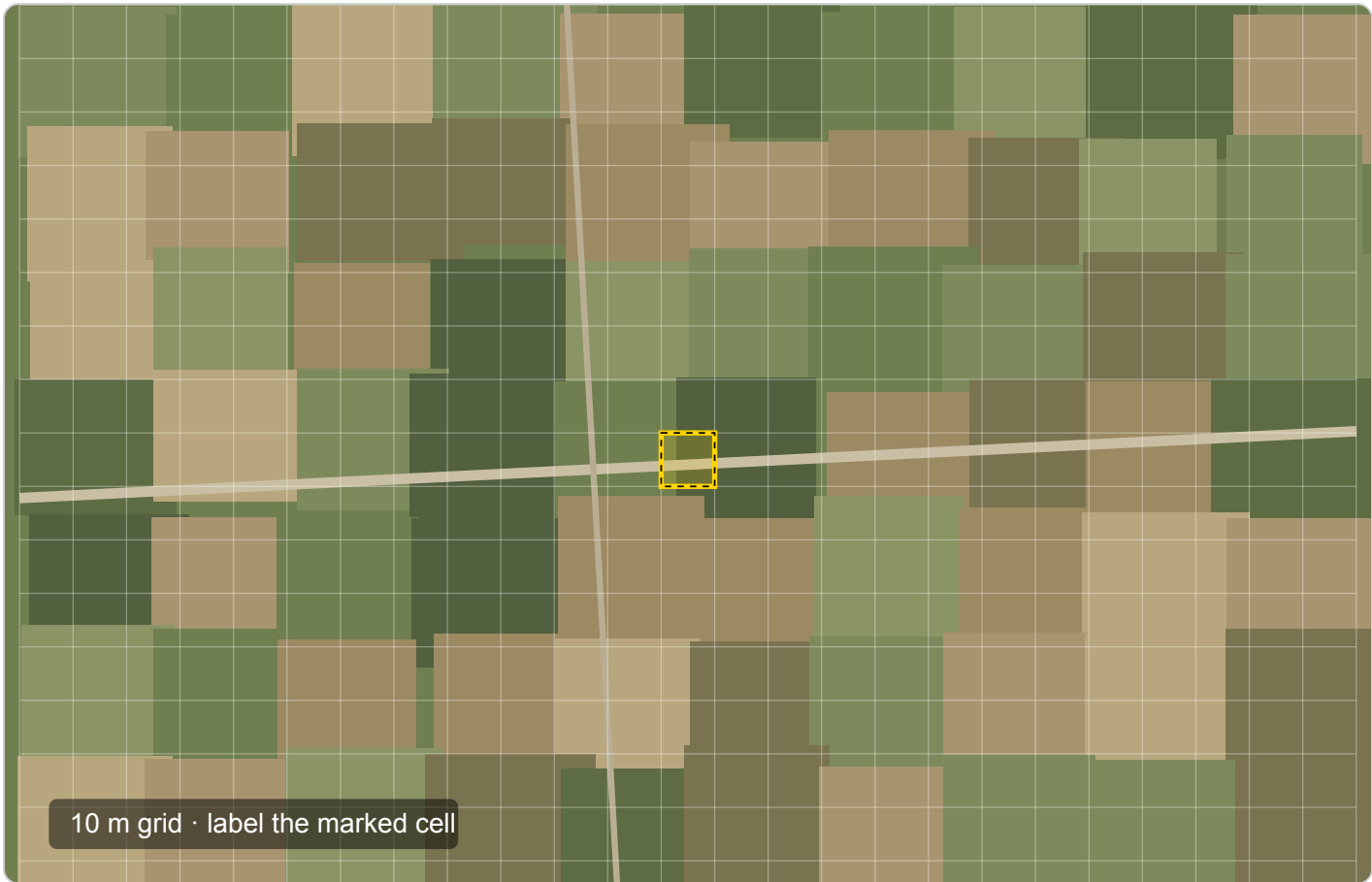
EPSG:21037 · schematic

A2 · Annotate

The same gate as N4, as progress. Adding labels can only improve the state, never worsen it.

LABEL THE CELL Pick the cover that occupies the whole 10 metre cell. Where more than one is present, choose what covers most of it — not the most interesting thing in it.

Location 284 of 800 0.4712 S, 37.0834 E High-resolution optical, 0.6 m · 2023



Zoom 100 m 250 m 500 m 1 km Imagery placeholder

Woody
1

Not woody
2

Every location returns a class. There is no unsure and no skip.

←

→

Label readiness

NOT ENOUGH

Fewer labels of at least one class than the recommendation.

Woody	121 / 400
Not woody	163 / 400
Smallest class	Woody, 279 short
Locations labelled	284 / 800
Blocks reached	279, one point each

Double-interpreted subset LATER
Measures interpreter error. Not in this release.

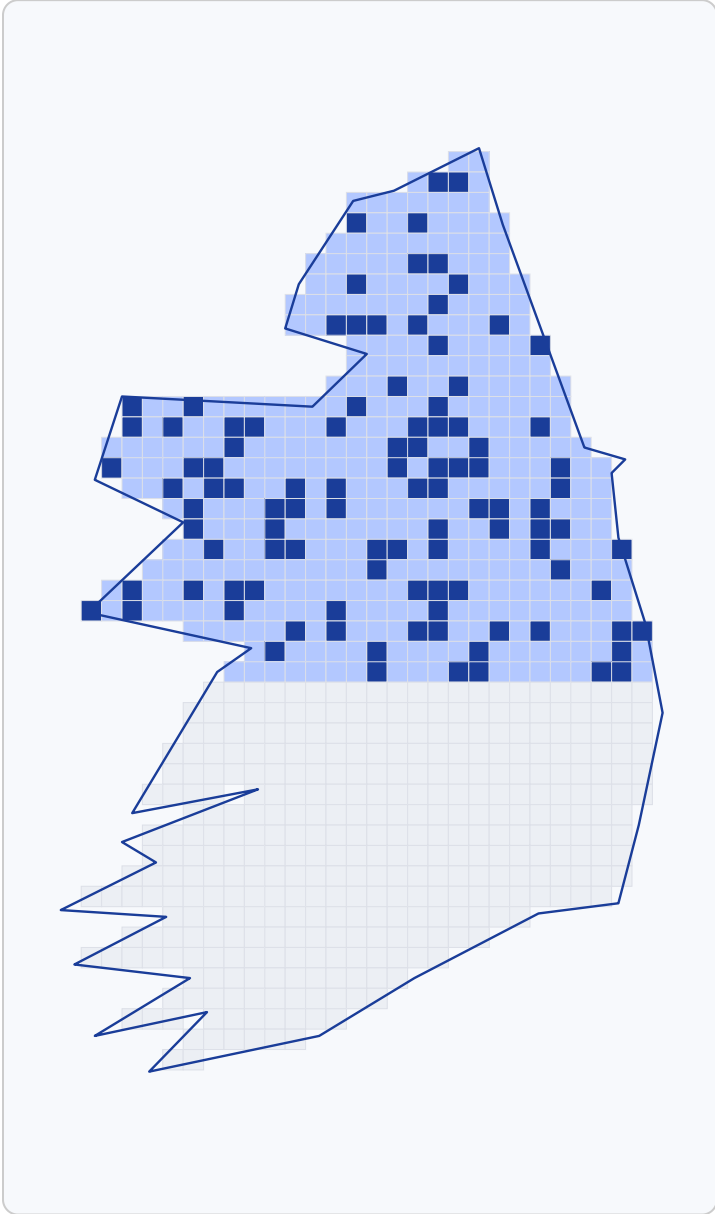
Continue — 279 more Woody needed

The smallest class decides. At 400 of each the run can go ahead; from 200 it proceeds with a caution.

N5 · Partition and run

Both branches converge. Block hold-out only, seed on screen, and one run control: the embedding year. Everything else is fixed and recorded.

What the model will never see



■ Training ■ Held out ■ No labels

Partition, then run

337

Training blocks

112

Held-out blocks, 25 percent

7,284

Training points

2,543

Held-out points

Blocks by stratum

Stratum is the share of positive points in the block. The draw itself is unstratified.

Stratum	Blocks	Training	Held out	Held-out points
Under 10 percent positive	198	149	49	1,004
10 to 40 percent positive	141	106	35	842
Above 40 percent positive	110	82	28	697
All blocks	449	337	112	2,543

Per-stratum split is illustrative. Totals are Ireland's real figures.

Embedding year

2018



The only control

Everything else is fixed and goes into the run record

200 trees · seed 42 · 25% block hold-out · 10 bootstrap draws, seeds 0 to 9 · 10 m cell, no neighbourhood mean · argmax class · Olofsson stratified estimator · 10 km UTM blocks, EPSG:2157

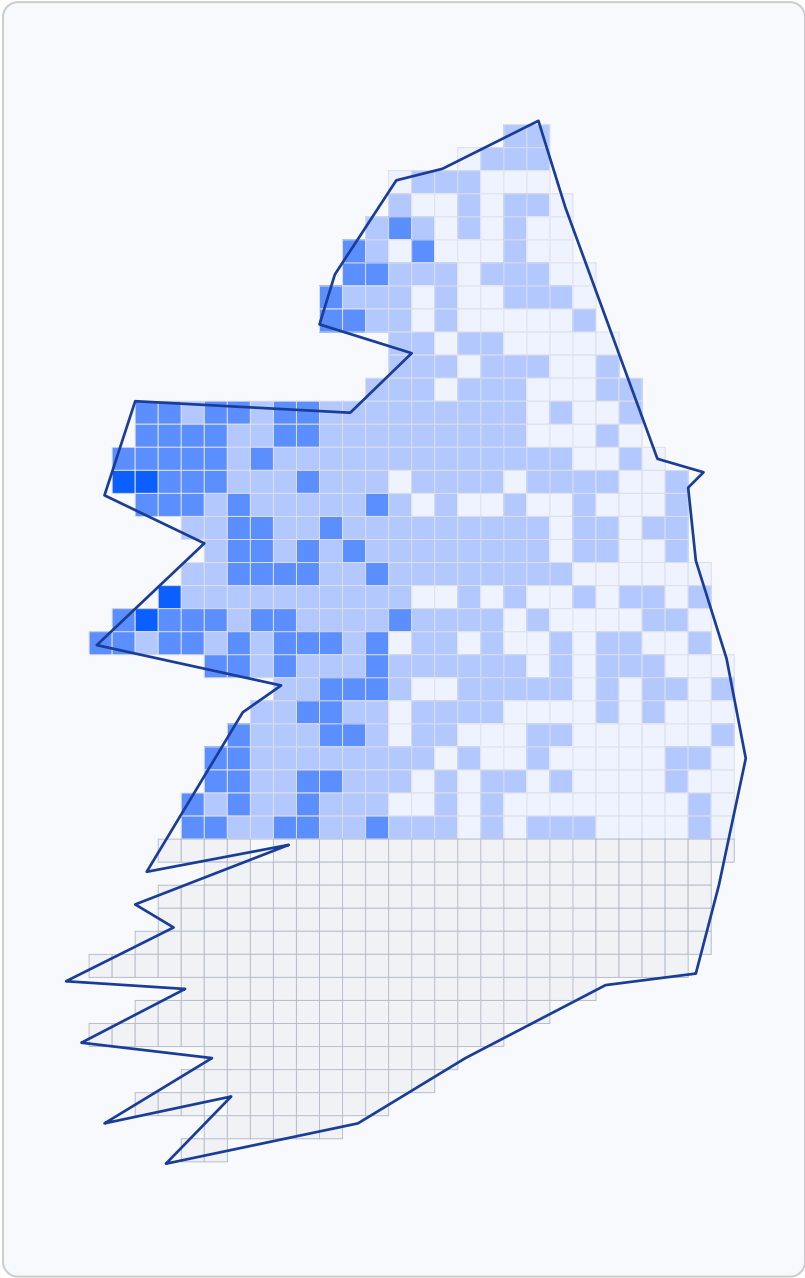
3 warnings carried. No position accuracy · dates 2011 to 2023 · no labels below 53.10°N

Submit job

Redraw with a new seed

seed 42

Prediction filling in



Peat probability 0 to 1 Pending

Run ie-peat-0142

Started 09:41:02

running

Job: ie-peat-0142 Blocks: 337 training, 112 held out Seed: 42 Elapsed: 3 min 12 s

09:41:02 Job accepted. 337 training blocks, 112 held out, seed 42.

09:41:02 Carrying 3 warnings from the readiness check into the run record.

09:41:05 Sampling embeddings at 9,827 points, 2018, 3 by 3 mean...

09:41:58 Embeddings sampled. 9,827 of 9,827 points returned a value.

N7 · Held-out accuracy

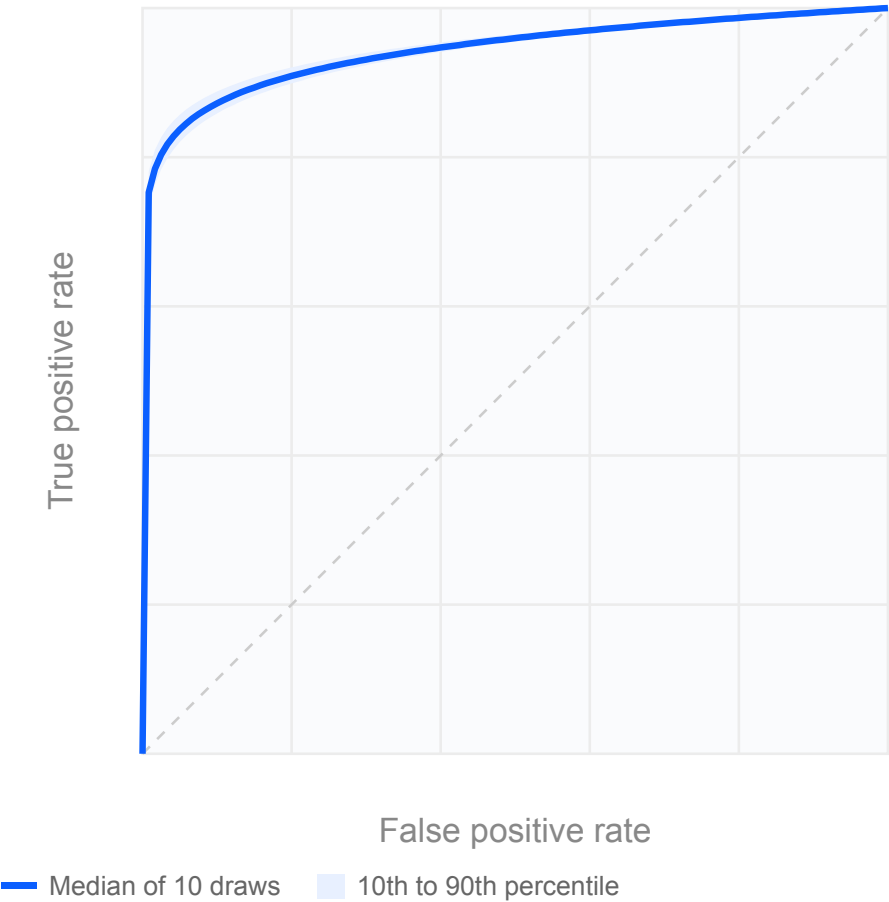
Median with a 10th to 90th band over 10 block-bootstrap draws. Categorical run shown; a continuous run reports R squared and root mean square error.

Accuracy on blocks the model never saw

2,543 points in 112 held-out blocks · median and 10th to 90th percentile over 10 draws

3 warnings carried

Receiver operating characteristic

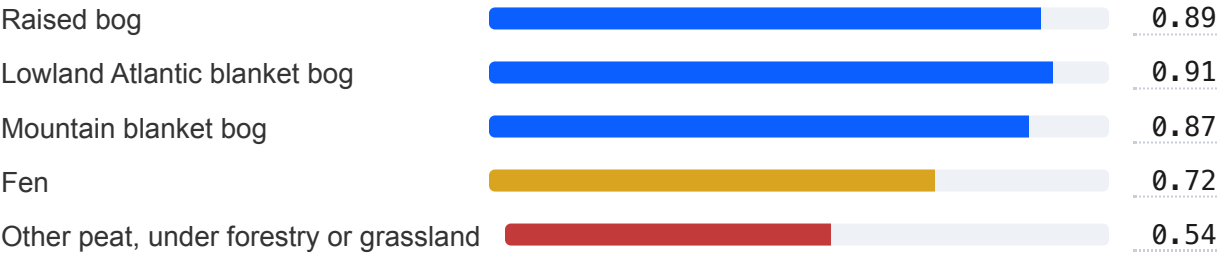


Held-out metrics

Measure	Median	10th to 90th
Area under the curve	0.944	0.938 to 0.951
F1, positive class	0.833	0.827 to 0.838
Overall accuracy	0.90	0.89 to 0.91
Producer accuracy	0.84	0.82 to 0.86
User accuracy	0.83	0.81 to 0.85
Area-share error	+2.6%	p90 err 2.6%

Dotted underline = placeholder

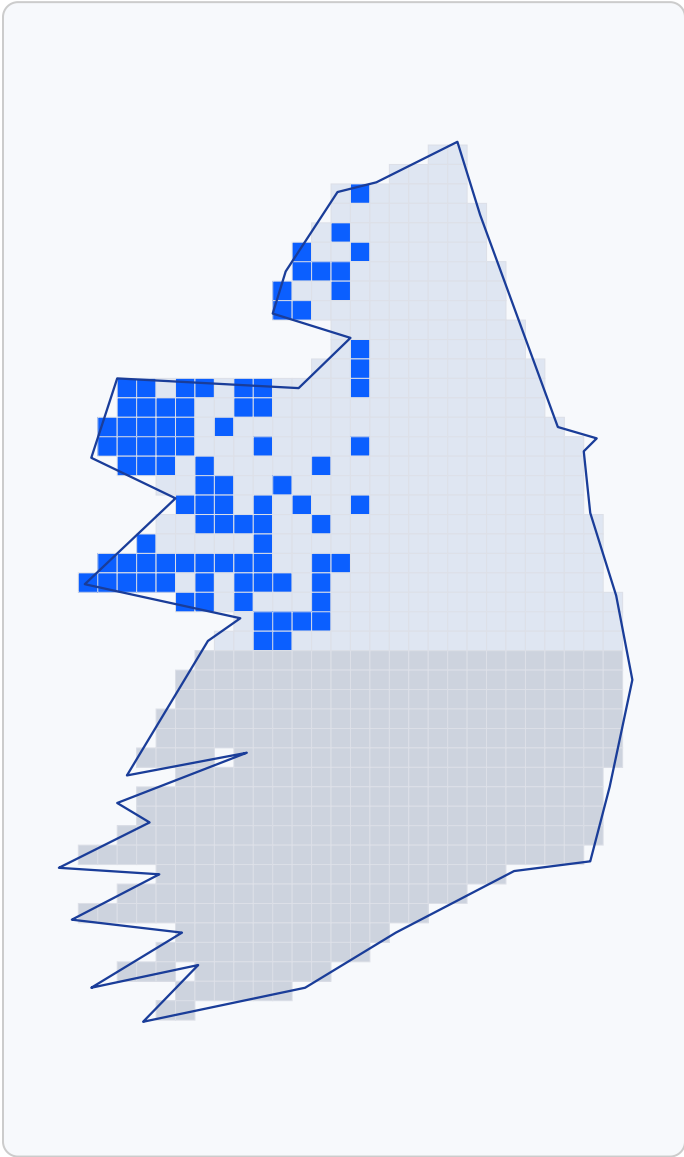
Where the model fails



N8 · Extent and its error

Where the tool stops. A stratified reference-based estimate with an interval, reported only over blocks that hold reference points.

Peat soil, 2018



■ Peat ■ Not peat ■ Masked

Extent and its error

3 warnings

Peat extent

1,321,722 ha

95 percent interval 1,268,464 to 1,374,981

Area-share error on held-out blocks

-13.1%

90th percentile 15.2 percent

Error apportionment

DOMINANT	Peat definition, set by the agency before upload	0.51
MATERIAL	Classification error on held-out blocks	0.28
MATERIAL	Blocks outside the labelled range	0.17
MATERIAL	Embedding year against the label years	0.11
NEGLIGIBLE	National boundary delineation	0.03

Next

Extent feeds your indicator calculation. Orbit stops here.

Masked

41 percent of blocks, below 53.10°N. No national total.

Take it away

Products

- [↓](#)**peat_probability_2018.tif**
HEADLINE. Class probability, 10 m, uint16 scaled 0-10000, tiled by 10 km block206 MB
- [↓](#)**extent.csv**
Extent with its 95 percent interval, computed at argmax2 KB
- [↓](#)**accuracy_by_draw.csv**
10 bootstrap draws, every metric18 KB
- [↓](#)**run_record.yaml**
Full run record6 KB

Download everything

Send to an Earth Engine asset

No hard-class raster. You threshold the probability. The published extent is computed at argmax and is the only figure with an interval.

3 warnings stamped on every product. No position accuracy · dates outside 2017 to 2024 · no labels below 53.10°N

Run record

run_record.yaml

```
run_id: ie-peat-0142
characteristic: peat soil      label_type: categorical
intended_use: "SDG 6.6.1, vegetated wetland extent" # recorded, not used
country: IRL  grid: 10 km, EPSG:2157
blocks_total: 449

upload:
  file: tellus_shallow_topsoil_A.csv
  sha256: 4f1c...9ab2  rows_read: 9,921
rows_used: 9,827
  crs_detected: EPSG:4326 (from the values and column names)
  columns: {lon: Longitude, lat: Latitude, label: PEAT_CLASS,
            date: SAMPLE_DATE, accuracy: null, site: SAMPLE_ID}
  classes: {Peat: positive, Not peat: absence}
  cleaning: 92 outside boundary, 0 duplicates at 10 m

gates:
  readiness: high (predicted interval 1 to 2 pp, per class)
  absence_declared: yes, on N3
  warn.position_accuracy_column: absent
  warn.observation_year: 2011-2023, outside 2017-2024
  warn.coverage: no labels below 53.10 N (41% of blocks masked)

partition:
  rule: block hold-out, unstratified draw
  seed: 42  test_blocks: 112 (2,543 pts)
  train_blocks: 337 (7,284 pts)

block_ids: see partition_blocks.csv

model:
  classifier: ee.Classifier.smileRandomForest
  trees: 200  seed: 42  features: AE 2018, 10 m cell
  class_rule: argmax_class (no threshold parameter)
  draws: 10, block bootstrap with replacement, seeds 0-9

results (median, 10th-90th percentile over draws):
  auc: 0.944 (0.938-0.951)  f1: 0.833
  overall_accuracy: 0.90
  area_share_error: +2.6% (p90 |err| 2.6%)

area:
  estimator: Olofsson 2014 eq. 9-11, at argmax
  extent_ha: 1,703,000  ci95: [1,610,000, 1,796,000]
  scope: the 449 blocks holding points, not a national total

export:
  task_id: 2026-08-22-irl-peat-0001
  tiles: 449 COGs, one per 10 km block
  band: under 2 pp, silent

versions:
  tasking_tool: national_tasking @ 2026.08.0
  embedding:
    GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL
    boundary: national high-resolution coastline (not GAUL 500 m)
```

BLOCKED · NO ABSENCE CLASS

This upload has no absence class

⚠ **5,006 plots, 98 percent of them carrying at least 10 percent canopy.** The inventory visits forest land only, so it cannot supply absences for a forest-or-not question. More plots of the same kind do not fix that.



Annotate the absences RECOMMENDED

Returns to the sampling-design screen in absence mode. We draw a sample away from your plots and you mark each one on imagery. Real absences, not manufactured ones.

Set up

Upload negatives

A second file of true negatives from the same survey programme, sampled by the same design.

Rarely exists. That is why you are on this screen.

Upload

Random background

Background drawn across the country, outside a 1 km buffer of your plots.

Some background falls on unmapped forest. Negatives contaminated at an unknown rate.

Use random background

Background from a mask

Background drawn only inside a map of where the class is not, which you supply.

If the mask is the map you compare against, the comparison is no longer independent.

Upload a mask

Two things stop a run: a missing absence class, and an interval in the low band. The finding and the ways out change; the layout does not.

No absence class	Annotate absences, upload negatives, random background, background from a mask	declared on N3
Interval low, one class	Annotate that class, upload more field data for it, merge it into a neighbour, or drop it and state the reduced set	over 5 pp
Interval low, every class	Add points, or reduce the country to an area of interest where coverage is sufficient	over 5 pp